

BTEC Level 3 Applied Science • Unit 1 Chemistry • Worksheet

Metallic Bonding

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

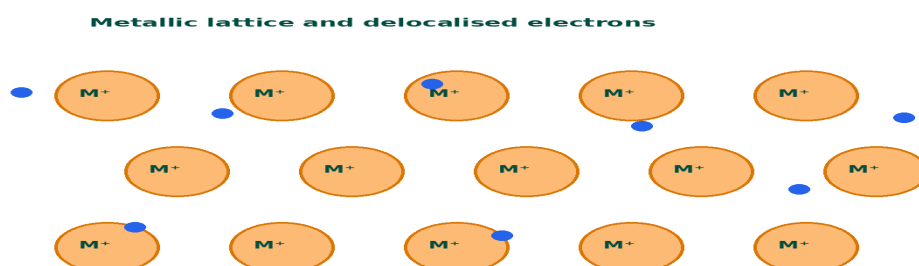
Learning focus: delocalised electrons, properties and comparison with ionic solids.

A Retrieval

1. Define metallic bonding.

2. Which particles move when a metal conducts electricity?

3. What is meant by malleable?



Positive metal ions in a sea of delocalised electrons. The attraction is metallic bonding.

4. Describe the structure of a solid metal using the diagram.

5. Explain why metals conduct heat well.

6. Explain why metals can be drawn into wires.

C Application

7. Why does solid magnesium conduct but solid magnesium oxide does not?

8. Suggest why steel is harder than pure iron.

9. Why are Group 1 metals softer than many transition metals?

D Exam-style practice

10. Describe the bonding in solid magnesium. [3]

11. Explain why metals are malleable but ionic crystals are brittle. [4]

12. Explain why copper is used for electrical wiring. [2]

E Challenge

13. Use ideas about charge and delocalised electrons to suggest why aluminium has a higher melting point than sodium.
